

Inhibitor Synergy in the bAID Host

What the 2021 Bioscreen runs measure, what Bliss and Loewe explain, and what a chemogenomic model trained on public screens adds

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Abstract

Six sorghum-hydrolysate inhibitors, their 63 combinations and three two-way dose grids were measured on one strain in 2021: bAID, the CRISPR-AID host of Lian 2019, carrying no deletion, in aerobic YPD, 977 wells over five Bioscreen C runs. Four claims were registered before anything was scored, and the model-free half of the analysis is now complete on all four.

Growth under the combinations departs from independence and stops short of Loewe additivity. Loewe from the single-agent Hill fits separates the combinations that grew from those that did not at AUROC 0.958 and accuracy 0.825 over all 63, while Bliss independence from those fits and highest single agent predict growth for every one of the 63 (accuracy 0.333, AUROC 0.777 and 0.663) and Bliss from the observed single-agent wells reaches 0.556 and 0.872; 42 of 63 did not grow within 85 h. On the 21 that grew, Bliss from the observed single-agent wells reaches Spearman 0.573 and overestimates fitness by 0.142 on average, Loewe reaches 0.713 and underestimates it by 0.128. Twelve of the 15 pairs are synergistic against Bliss with bootstrap intervals below zero, two are additive and one antagonistic, and no interior cell of any isobole grid grew less than Loewe predicts. Every score is reported under both growth calls, which disagree on 12 of 197 ex23 wells, all of them 5-HMF combinations.

What the chemogenomic representation adds is, so far, nothing measurable. Profile similarity does not anticipate which pairs deviate: with the measured profiles in the matrix no similarity measure reaches $p = 0.26$ over 15 pairs, and the only association runs the other way in the all-predicted matrix, which is fingerprint similarity among the small acids. Compound-cold ridge profiles reach a median centered Spearman of 0.387 over 32 screened compounds and carry only the generic stress component: 19 of 20 significant terms for predicted furfural are terms of the 32-compound gene mean, and no compound shares a significant term between its measured and predicted centered profiles. On public data the composition rule is measurable and is the mean, not the sum: over Hillenmeyer's 26 two-compound environments the mean of the matched single profiles has the highest R^2 of the fixed rules in 24 of 26 pairs (median 0.365 against 0.091), with the per-pair linear fit giving $a + b = 1.06$, while the emergent and masked gene rates sit inside the hit-calling noise floor. The bars a dose- and mixture-aware model has to clear are therefore model-free and already on file; the trainer that would clear them is being written and nothing has run.

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1 The question, and what the 2021 runs measured x

A chemogenomic screen measures a compound’s effect on a pool of deletion strains, and the models of experiments 031 to 038 predict that effect from the compound’s structure. The host those models never see is the one a fermentation runs in: an engineered strain with no deletion, growing in a medium that holds several inhibitors at once. The question is whether a model fit on public chemogenomic screens can say anything about that host, and the measurement that can answer it already exists.

bAID is the strain of every run below. Lian 2019 built it by integrating PmeI-digested pAID6 into BY4742 and selecting for G418 resistance, and its Supplementary Table 9 gives the genotype as BY4742-Delta::KanMX-[dLbCpf1-VP]-[Csy4]-[dSpCas9-RD1152]-[SaCas9]; the 2021 report calls the same strain BY4742-iAID6. ▲ [external: Lian 2019 (MinerU OCR of the publisher PDF, library key `lianMultifunctionalGenomewideCRISPR2019`); the sha256-pinned quotes are in `torchcell/datasets/private\_torchcell/volk2021\_sources.py`] It carries the CRISPR-AID machinery and no deletion, so a model asked about it is asked about the **empty genotype**: the wild-type genome under a perturbed environment.

The six inhibitors. Furfural, acetic acid, 5-(hydroxymethyl)furfural (5-HMF), formic acid, levulinic acid and lactic acid, the degradation products a bioenergy sorghum hydrolysate carries into a fermentation. Short forms FF, AA, HMF, FA, LVA and LA appear in the figure panels and nowhere in the prose.

The five runs. All on a Bioscreen C, 200 wells per run, optical density over time, aerobic YPD, biological replicates randomized across well positions (Table 1). The single-agent titrations (ex21) ran 72 h, the 63 combinations (ex23) 85 h, and the three two-way dose grids (ex26 to ex28) 96 h. The doses of ex23 were chosen from ex21 by the report’s own rule, the first concentration that showed an increase in doubling time while also showing low doubling time variance. ▲ [external: 2021 preliminary exam report, section 2.2.1 (MinerU 2.7.6 OCR, 200 dpi, of the submitted PDF; sha256 recorded in `results/wetlab\_provenance.json`)]

Table 1: The five Bioscreen C runs of 2021, as the served records carry them. Length is the longest well duration in the run, so a combination that never grew is a statement about that length and no longer. Growth is the served raw-curve call of Section 2. The isobole runs carry two biological replicates per dose cell rather than three.

Run	Design	Length (h)	Wells (n)	Grew (n)	Inhibitors (n)	Replicates (n)
ex21	single-agent titration, six inhibitors	72.0	180	117	6	3
ex23	combinations of one to six inhibitors	85.0	197	69	6	3
ex26	isobole grid, furfural x acetic acid	96.0	200	45	2	2
ex27	isobole grid, formic acid x acetic acid	96.0	200	76	2	2
ex28	isobole grid, 5-HMF x acetic acid	96.0	200	53	2	2
all			977	360	6	

The four claims, as registered. Each was allowed to fail, and the order was fixed: the model-free references first, because a reference rule that already explains the combinations leaves a trained model nothing to add.

1. Growth under combinations departs from independence. Reference rules from the ex21 single-agent curves: Bliss independence, Loewe additivity, highest single agent; scored on the 63 ex23 combinations (grew within 85 h over 63; fitness over the 25 that grew) and the three isobole grids (front against the Loewe line). Hypothesis: synergistic relative to Bliss (Bliss gives 0.60 for the four mildest singles together; nothing with four grew).
2. Chemogenomic profile similarity anticipates which pairs deviate. Measured Vanacloig profiles where they exist, compound-cold ridge-FCFP4 profiles for formic and lactic acid (and, as the fairness check, for the shared compounds from their held-out folds). Tested on 15 pairs (permutation) and the three isoboles; HMF x acetic acid is shown but excluded (single run, growth islands; 039).
3. Deletion profiles compose under a mixture by a rule measurable on public data: Hillenmeyer HET’s 26 two-compound environments against their single-compound partners (sum, mean, max, Bliss-style product); the base rate of emergent and masked genes.
4. Post-analysis only, not a validated prediction: composed profiles nominate the genes a screen under each combination would recover (core, compound-specific, conflict). The check is GO enrichment of the nominated sets, and whether it overlaps the terms the measured single-compound profiles carry. No in-house screen exists to test them (the thesis archive holds only the library-sequencing correspondence for that screen).

Measured. All four claims on CPU, from the committed scripts of `experiments/040-inhibitor-synergy-wetlab/scripts/`: claim 1 holds, claim 2 fails as registered, claim 3 holds on public data, claim 4 is a post-analysis whose nominations are hypotheses. Every score is reported under both growth calls.

Not measured. Any trained model on this task. The mixture trainer is being written, no run exists, and no number below comes from one. No screen of the bAID deletion pool under any inhibitor combination exists, so claim 3’s composition rule is carried to these compounds as a labeled hypothesis and claim 4’s nominations are untested.

Vocabulary. **Fitness** is the ratio of the uninhibited doubling time to the well’s doubling time, so the uninhibited control is 1 and a well that never grew is 0. The **served call** derives that doubling time from the raw optical-density curve and is primary; the **software call** takes the Bioscreen software’s generation time and is carried as a sensitivity (Section 2). A **combination** is a set of inhibitors each at one dose, and the 63 combinations of ex23 are every non-empty subset of the six that the design covers. A **profile** is a compound’s vector of per-gene deletion responses, $\log_2$ of treatment over control, negative when the deletion is sick. **Compound-cold** means a compound whose profile was held out of the fit entirely, which is what a prediction for a never-screened inhibitor has to be. A **reference rule** predicts a combination’s fitness from its single agents and contains no fitted parameter beyond the single-agent curves.

2 Data: one private strain screen and four public chemogenomic screens ✗

The private dataset. `InhibitorBioscreenVolk2021Dataset` serves 977 records, one per well, from the dev LMDB at `\$DATA\_ROOT\data/torchcell/inhibitor\_bioscreen\_volk2021`. Every record carries the run, the well duration, each inhibitor’s dose, the derived fitness and the growth call. The provenance record (`results/wetlab\_provenance.json`, pydantic `WetlabProvenance`) pins the report OCR by sha256 together with the raw-mirror manifest and the eight archive files the loader consumed, so the table below rebuilds from hash-pinned bytes rather than from a directory listing. The tidy well table (`results/wetlab\_wells.csv`) is read from that LMDB and is re-pointed at the knowledge graph once the served build carries the private dataset; the records are the same.

The growth call, and why there are two. The served records call growth from the raw curve: a well grew when its baseline-subtracted optical density rose by at least 0.3, and its generation time is taken from the steepest 3 h window. The 2021 analysis instead used the Bioscreen software’s generation times, computed by PRECOG from calibrated cell counts. ▲ [external: 2021 report, section 1.2.3: Growth curves are then processed with PRECOG software using estimated cell counts from calibration curves to compute doubling times] The two calls agree on 185 of 197 ex23 wells, which the loader pins at 0.939086. The 12 that disagree are all three replicates of four 5-HMF combinations: 5-HMF with lactic acid, 5-HMF with levulinic acid, furfural with acetic acid and 5-HMF, and furfural with 5-HMF and formic acid. The software assigned them generation times of 6.6 to 10.6 h while their optical density rose only 0.085 to 0.277, below the 0.3 rise the raw-curve call requires (`results/wetlab\_growth\_call\_check.csv`). The scales also differ: the ex23 control generation time is 1.345 h from the raw curves and 1.815 h from the software ($n = 8$ control wells), so the two fitness columns are never mixed.

The served call is primary. The disputed wells are all 5-HMF combinations, which is exactly where a synergy call for 5-HMF would land, so the software call is carried as a sensitivity and every claim-1 score is reported under both.

Table 2: The 63 ex23 combinations under the two growth calls. A combination grew when at least one of its three wells grew; the mean fitness is over the combinations that grew, each on its own call’s scale. The software call reproduces the counts of experiment 039 exactly, which places the difference in the call rather than in the table.

Inhibitors (n)	Combinations (n)	Grew, served (n)	Grew, software (n)	Mean fitness of grown, served	Mean fitness of grown, software
1	6	6	6	0.805	0.788
2	15	12	14	0.437	0.465
3	20	3	5	0.426	0.461
4 to 6	22	0	0		
all	63	21	25		

Doses. Every dose is stored in g L^{-1} , the unit the stock sheets use, and converted to mM with the standard formula weight, which is what makes a dose comparable to a public screen’s (Table 3). The spread is wide: at the ex23 doses lactic acid is 222 mM and furfural is 15.6 mM, a factor of 14, so a rule that ignores potency and counts inhibitors cannot order these combinations.

The public tables, and which inhibitor each screen holds. The pooled chemogenomic table of experiment 033 (build 002, TMM-normalized) holds the 32 published Vanacloig-Pedros 2022 compounds `10.1093/femsyr/foac036`, of

which only furfural and 5-HMF are among the six. Levulinic acid and sodium acetate were dropped from build 002 as tokens the paper does not report (issue #501); build 001 still serves them and is read for supplementary comparison rows only. Sodium acetate is measured in Hoepfner 2014 10.1016/j.micres.2013.11.004, at 125 and 150 mM in YPD with homozygous diploid deletions, a different screen. Formic and lactic acid are in no screen, so their profiles can only be predictions. Hillenmeyer 2008 10.1126/science.1150021 contributes no inhibitor, and enters as the only public data with two compounds in one well (Section 3).

Table 3: The six inhibitors: dose at ex23, public coverage, and the profile the best-available matrix uses for each. Replicate reliability is the Spearman correlation between the screen’s own replicate profiles, and the column is blank where the source screen gives no replicate pair. Acetic acid is matched to Hoepfner’s sodium acetate, the same anion at YPD pH, which is a stated assumption and not a measurement; 5-HMF is measured but its reliability is below zero, so the matrix uses its prediction instead.

Inhibitor	Short	M (g mol ⁻¹)	ex23 dose (g L ⁻¹)	ex23 dose (mM)	Vanacloig store	Replicate reliability (Spearman)	Profile used
furfural	FF	96.08	1.50	15.6	furfural, b002	0.769	measured, build 002
acetic acid	AA	60.05	2.00	33.3	sodium acetate, b001		measured, Hoepfner
5-HMF	HMF	126.11	2.52	20.0	5-HMF, b002	-0.048	predicted, held out
formic acid	FA	46.03	1.00	21.7	none		predicted, on 32
levulinic acid	LVA	116.12	6.00	51.7	levulinic acid, b001		predicted, on 32
lactic acid	LA	90.08	20.00	222.0	none		predicted, on 32

Confounds, carried into every claim. Three, and none is removable from these data.

- **Run length.** 72 h for ex21, 85 h for ex23, 96 h for the isoboles. No growth is a statement about the run, so a combination recorded as not grown in ex23 might grow in 96 h. The classification scores of claim 1 inherit this.
- **Strain and medium.** bAID in aerobic YPD against Vanacloig’s haploid deletion pool in anaerobic SYNH3 at IC30, and against Hillenmeyer’s heterozygous diploid pool in aerobic YPD read as log₂ depletion over 20 generations. A profile measured in one is carried to the other without a correction.
- **The ex21 control.** Its uninhibited wells grew more slowly than its low-dose wells (mean fitness 1.012, 17 of 18 grew), so ex21 is normalized by the fitted top of each Hill curve rather than by its controls, and the fitted tops run 1.00 to 1.25.

3 Methods ✗

3.1 Single-agent curves ✗

Index the six inhibitors by i and write d_i for the dose of inhibitor i in a well, in g L⁻¹. The fitness of a well holding inhibitor i alone is modeled by a Hill curve with three parameters, a top θ_i , a half-maximal dose c_i and a slope h_i ,

$$f_i(d) = \frac{\theta_i}{1 + (d/c_i)^{h_i}}, \quad \varphi_i(d) = \frac{f_i(d)}{\theta_i} = \frac{1}{1 + (d/c_i)^{h_i}}, \quad (1)$$

where $\varphi_i(d) \in (0, 1)$ is the **fractional fitness**, the fitness relative to the curve’s own top. Fitting the top rather than dividing by the controls is what the ex21 control anomaly of Section 2 forces. Inverting Equation 1 gives the dose of inhibitor i alone that leaves fractional fitness y ,

$$D_i(y) = c_i \left(\frac{1 - y}{y} \right)^{1/h_i}, \quad 0 < y < 1, \quad (2)$$

so $c_i = D_i(1/2)$ is the IC50 and $D_i(0.7)$ is the IC30. Parameters are fit by least squares on the dosed wells of one run, with h_i bounded above at 20. The primary variant puts a well that did not grow at fitness 0 and the **excluded** variant drops it; intervals are the 2.5 and 97.5 percentiles of 1000 resamples of the replicate wells within each dose. Two properties of the fits matter downstream. A slope at its bound, $h_i = 20$, marks a cliff, fitness falling from near the top to none between two adjacent tested doses, so c_i is located only within that dose interval and the bootstrap understates it, since resampling cannot move a cliff that sits between doses. A fitted c_i at or above the highest dose tested is an extrapolation of the curve rather than a reading of it.

3.2 The three reference rules ✖

Let S be the set of inhibitors present in a combination, each at its own dose d_i , and let E denote a predicted fractional fitness for that combination. Every rule below takes the single-agent curves of Section 3.1 and no fitted parameter of its own.

Bliss independence. Bliss 1939 treats the agents as independent events acting on survival, so the surviving fraction multiplies,

$$E_S^{\text{Bliss}} = \prod_{i \in S} \varphi_i(d_i), \quad (3)$$

which for a pair A and B is the familiar $f_{AB} = f_A f_B$. Two variants differ only in where $\varphi_i(d_i)$ comes from: `bliss_ex21` reads it off the ex21 Hill fit at the combination’s dose, and `bliss_ex23` uses the observed ex23 single-agent wells at that dose, the mean of three wells with a well that did not grow counted as 0. The second is the stronger reference, since it carries the run’s own singles rather than another run’s curve.

Loewe additivity. Loewe additivity asks what dose of each agent alone would produce the combination’s effect, and calls the agents additive when those doses sum to one full dose. With D_i from Equation 2, the prediction is the effect E solving

$$\sum_{i \in S} \frac{d_i}{D_i(E)} = 1. \quad (4)$$

The left side is monotone in E , because $D_i(E)$ grows without bound as $E \rightarrow 0$ and vanishes as $E \rightarrow 1$, so the root is unique and is found numerically on logit E by bisection. Evaluated at an observed fractional fitness y instead, the same sum is the **combination index**

$$\text{CI}(y) = \sum_{i \in S} \frac{d_i}{D_i(y)}, \quad (5)$$

below 1 for synergy and above 1 for antagonism relative to Loewe. A component whose $D_i(E)$ lies above the highest dose ex21 tested is flagged: that single never reaches E within its measured range, so its term is a Hill extrapolation.

Highest single agent. $E_S^{\text{HSA}} = \min_{i \in S} \varphi_i(d_i)$, the sickest single agent, in the ex21-fit and observed-ex23 variants.

3.3 Scoring the combinations ✖

The growth threshold. A rule predicts fitness, and a well is recorded as grown or not, so the two are joined by a threshold τ taken from the data: the lowest fitness of any grown inhibited ex23 well under that call, which is the slowest growth the call registers within 85 h. A rule predicts growth when its prediction is at least τ . The thresholds are 0.248 for the served call and 0.171 for the software call. Classification is scored as accuracy and as AUROC with the rule’s prediction as the score, so AUROC reads the ordering and is free of τ .

Fitness of the grown combinations. Observed fitness is the mean of the grown wells, the definition experiment 039 used. Rules are scored by Spearman correlation, by RMSE and by the mean of observed minus predicted, whose sign says which way a rule errs, over all 63 combinations and over the 57 holding two or more inhibitors; the rows by number of inhibitors are in `results/mixture\_scores.csv`.

Pair deviations. For each of the 15 pairs, the deviation is observed fitness minus E^{Bliss} from the ex23 singles, with a 95 percent interval from 2000 resamples of the replicate wells of the pair and of both singles (and of the Hill fit draws for the ex21 variant). A pair is called synergistic when that interval lies below zero, antagonistic when it lies above, and additive otherwise.

Isoboles. Each grid holds two inhibitors over nine doses each, two plate wells per cell. The Bliss surface is built empirically from the grid’s own axes, so the excess over Bliss is a within-run quantity, averaged over the 81 interior cells with a 95 percent interval from 2000 resamples of the two wells per cell. The Loewe additive front is the locus of Equation 4 at $E = \tau$ computed from the grid-axis Hill fits, and the cells that contradict it are counted in both directions.

3.4 How deletion profiles compose, measured on public data ✖

Claim 3 cannot be asked of the inhibitors, because no screen of a deletion pool under any inhibitor combination exists. Hillenmeyer 2008’s heterozygous diploid screen (HET) is the one public source with two compounds in one well: of its 452 environments in the 033 pooled table, 26 carry two compounds, over 5,825 genes. The response of gene g in environment e is $y_{ge} = \log_2(\text{mean control intensity}/\text{treatment intensity})$, so a positive value is a fitness defect; where

the 033 flatten keeps several measurements for a cell, their mean is used. Each two-compound environment is matched to the single-compound environment of each partner at the same $\log_{10}$ molar dose, else the nearest dose with the same generation count. A match is reasonable when the dose is within one twofold step and the generations agree: 46 matches are exact, 2 within twofold, and 4 are beyond, every one of those four a tacrolimus pair whose dosed singles sit 402 to 804-fold away. Those four are flagged and reported separately.

Writing y_1 and y_2 for the two matched single profiles and y_{12} for the pair’s, four fixed rules and one fitted one are scored on the genes measured in all three environments: the sum $y_1 + y_2$, the mean $(y_1 + y_2)/2$, the elementwise maximum, the Bliss product on the fitness scale, and a free linear fit $y_{12} = ay_1 + by_2 + c$. On this scale the Bliss product is the sum,

$$-\log_2(2^{-y_1} 2^{-y_2}) = y_1 + y_2, \quad (6)$$

and the script confirms it: the largest absolute difference between the two is 0 on all 26 pairs, so the Bliss column carries no separate information and is plotted as the sum. Fixed rules are scored by $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ as they stand, with no refit, so a rule of the wrong magnitude scores negative; Spearman is scale-free, which is why the sum and the mean share one Spearman value.

A **hit** is a gene with robust $|z| > 2$ within its environment, where $z = (y - \text{median})/(1.4826 \text{ MAD})$ over that environment’s genes; 10.2 percent of all gene-by-environment cells are hits. An **emergent** gene is a hit in the pair and in neither single, a **masked** gene a hit in a single and not in the pair; the emergent fraction divides by the pair’s hits, the masked fraction by the union of the singles’ hits. Two nulls of 200 draws each replace the matched singles by random ones: any two YPD singles without either partner compound, and the stricter null restricted to singles sharing a control set with the pair, which all 26 pairs do. The noise floor is the only near-replicate pair of single environments in the release, two 5-fluorouracil doses within 1 percent of each other.

3.5 Compound-cold profiles, similarity, and GO enrichment ✖

The predictor. Experiment 038’s nested ridge on FCFP4 counts, the model that sets the bar on the corrected store. The fingerprint is the 031 recipe: RDKit Morgan with radius 2 over 2048 bits, feature atom invariants, counts rather than bits, float32, from the curated identity table’s SMILES. Re-featurizing 35 compounds reproduced every 031 stored row exactly (`results/profile\_fingerprint\_check.csv`), which is what makes a prediction for a compound outside the 32 comparable to one inside. Formic acid (OC=O) and lactic acid (CC(O)C(=O)O, no stereocenter stated on the stock sheet) are identity gaps in the private loader and are assigned here.

The regularization strength, rank and clip are chosen by leave-one-compound-out inside the fitted pool on the centered target. Furfural and 5-HMF are predicted with themselves held out entirely (31 fitted); the other four are predicted from a fit on all 32. A prediction is scored by Spearman correlation against the measured profile, raw and after centering each gene by its mean over the 32 compounds, which removes the component every compound shares.

Similarity. Two measures per compound pair, each raw and centered: the Spearman correlation over the genes both profiles carry, and the Jaccard overlap of the 100 most sensitive genes. Claim 2 tests each against the pair deviation of Section 3.3 over the 15 pairs with a 10,000-draw permutation p and a leave-one-pair-out range, under both growth calls and both profile matrices: the **best-available** matrix, which uses a measured profile wherever its replicate reliability reaches 0.5, and the **all-predicted** matrix, in which every column is a prediction. The 0.5 reliability threshold is a choice, not a sourced value.

GO enrichment. Gene-to-term annotations come from the S288C GFF `Ontology_term` fields of the torchcell genome, propagated to `is_a` ancestors through the GO release of 2024-01-17 whose sha256 is recorded in `results/go\_summary.json`. The background is the 3,587 build-002 Vanacloig genes, all annotated, and the 2,292 terms with 5 to 500 background genes are tested one-sided hypergeometric with Benjamini-Hochberg control per set at FDR 0.05. Study sets are the 100 most sensitive genes of each measured and predicted profile, the core nomination set, and each combination’s composed top 100. The control set is the 100 most sensitive genes of the build-002 gene mean over the 32 compounds, which is what a model with no compound-specific signal outputs, so a term the prediction shares with the control is evidence of the generic stress component rather than of the compound.

4 Results x

4.1 Claim 1 holds: the combinations grow less than independence predicts and more than Loewe additivity predicts x

The single agents. The ex21 titrations give a Hill fit per inhibitor over 27 wells each (Table 4, Figure 1). Potency spans a factor of 50 on a molar basis, from furfural at IC₃₀ 9.8 mM to lactic acid at 486 mM, and the curves are steep: formic acid and lactic acid both sit at the slope bound $h = 20$, as do most isobole axes, so for those compounds fitness falls from near the top to none between two adjacent doses. Levulinic acid’s fitted IC₅₀ of 10.2 g L⁻¹ lands on the highest dose tested, 10 g L⁻¹, so its curve is extrapolated above that point.

Against the ex23 singles at the ex23 doses, the ex21 fit agrees within the 95 percent interval for furfural (+0.03), acetic acid (+0.02), formic acid (0.00) and lactic acid (-0.02), and reads high for 5-HMF (-0.09, [-0.12, -0.05]) and levulinic acid (-0.09, [-0.14, -0.04]), each over three wells (`results/single\_agent\_ex23\_check.csv`). Acetic acid, dosed in all four runs, gives IC₅₀ 4.00 g L⁻¹ in ex21 against 2.51, 2.90 and 2.94 in ex26 to ex28, and ex21’s interval [3.30, 4.37] overlaps none of the three isobole intervals. The runs differ in length and in the ex21 control anomaly, and these data do not identify which produces the offset.

Table 4: The ex21 single-agent Hill fits, primary variant (a well that did not grow enters at fitness 0). The fitted top is the curve’s own uninhibited asymptote, used in place of the ex21 controls. A slope at 20 is at its upper bound and marks a cliff between two adjacent doses, so the IC₅₀ interval there understates the uncertainty. Intervals are 95 percent percentiles of 1000 resamples of replicate wells within each dose.

Inhibitor	Wells (n)	No growth (n)	Fitted top (fitness)	Hill slope h	IC ₅₀ (g L ⁻¹)	IC ₅₀ 95% interval (g L ⁻¹)	IC ₃₀ (g L ⁻¹)	IC ₃₀ (mM)	IC ₃₀ 95% interval (mM)
furfural	27	16	1.187	2.05	1.43	[1.38, 1.52]	0.94	9.8	[9.1, 12.2]
acetic acid	27	12	1.217	5.29	4.00	[3.30, 4.37]	3.41	56.7	[47.3, 64.6]
5-HMF	27	12	1.002	5.16	2.66	[2.59, 2.70]	2.26	17.9	[17.2, 18.4]
formic acid	27	9	1.152	20.00 (bound)	2.10	[2.07, 2.14]	2.02	43.8	[43.1, 44.6]
levulinic acid	27	1	1.185	5.78	10.16	[8.61, 16.21]	8.78	75.6	[70.4, 100.0]
lactic acid	27	12	1.252	20.00 (bound)	45.70	[44.93, 46.72]	43.81	486.3	[478.0, 497.1]

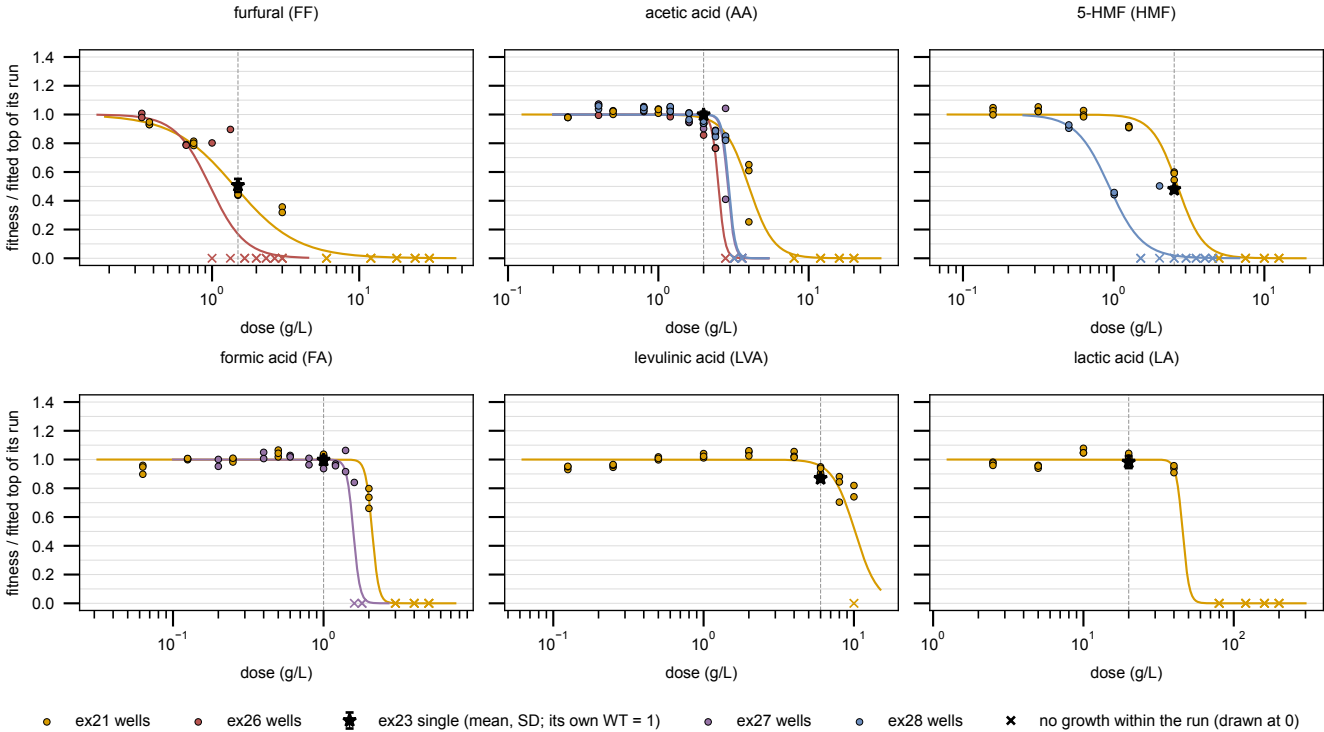


Figure 1: Each inhibitor’s dose response, per run, with the ex23 dose marked. Wells are divided by their own source’s fitted top and drawn with that source’s Hill curve; crosses are wells that did not grow within the run, placed at 0. Stars are the ex23 single-agent wells, mean and standard deviation of three wells on the ex23 scale where the uninhibited control is 1. The dashed vertical line is the ex23 dose, which is where every combination in Section 4.1 sits. The steepness of the acid panels is what separates the Bliss and Loewe predictions below.

Potency against the public screen. Only furfural and 5-HMF have a dose in the corrected Vanacloig store, and the

comparison is cross-medium rather than a replication: Vanacloig dosed the haploid pool at IC30 in anaerobic SYNTH3 read at 48 h, the Bioscreen ran bAID in aerobic YPD to 72 h. Furfural matches within a factor of 1.2, 5-HMF differs by 5.4 (Table 5). That is the resolution any dose-aware model inherits from those anchors alone.

Table 5: IC30 in the public screen against IC30 from the ex21 Hill fit. Levulinic acid has no condition in the corrected store, which the script confirms by reading all 34 condition blocks. Acetate is present only in build 001 and carries no IC30 row here.

Inhibitor	Vanacloig IC30 (mM)	ex21 IC30 (mM)	ex21 95% interval (mM)	Ratio ex21 / Vanacloig
furfural	8.0	9.8	[9.1, 12.2]	1.23
5-HMF	3.3	17.9	[17.2, 18.4]	5.42
levulinic acid	not in store	75.6	[70.4, 100.0]	

The 63 combinations. Growth is the discriminating readout: 42 of the 63 combinations did not grow within 85 h under the served call, 38 under the software call, and nothing holding four or more inhibitors grew under either. Loewe additivity reproduces that pattern and the other two rules do not (Table 6). Loewe reaches accuracy 0.825 and AUROC 0.958 served, and 0.825 and 0.952 under the software call, with one false positive in both. Bliss from the ex23 singles reaches 0.556 and 0.872 served; Bliss from the ex21 fits and both highest-single-agent variants predict growth for all 63 combinations, which is accuracy 0.333. Restricted to the 57 combinations of two or more inhibitors, Loewe holds AUROC 0.944 served and 0.939 software against Bliss from the ex23 singles at 0.844 and 0.817.

On the combinations that did grow, the two rules err in opposite directions (Table 7, Figure 2). Bliss from the ex23 singles reaches Spearman 0.573 served and 0.865 software and overestimates fitness by 0.142 and 0.097 on average; Loewe reaches 0.713 and 0.646 and underestimates it by 0.128 and 0.185. The samples are 21 and 25 grown combinations, and the three-inhibitor rows rest on 3 and 5 of them, too few to read. Loewe’s accuracy is also bought partly on extrapolation: for 33 of 63 combinations its solution lies beyond at least one component’s tested ex21 range.

Table 6: Growth or no growth over all 63 ex23 combinations. A rule predicts growth when its prediction is at least τ , the lowest fitness of any grown inhibited well under that call; AUROC uses the prediction as a score and is free of τ . Positives are 21 combinations under the served call and 25 under the software call. Best accuracy and best AUROC within each call in bold.

Reference rule	Growth call	τ (fitness)	TP (n)	FP (n)	TN (n)	FN (n)	Accuracy (fraction)	AUROC
Bliss, ex21 fits	served	0.248	21	42	0	0	0.333	0.777
Bliss, ex23 singles	served	0.248	20	27	15	1	0.556	0.872
Loewe, ex21 fits	served	0.248	11	1	41	10	0.825	0.958
Highest single agent, ex21 fits	served	0.248	21	42	0	0	0.333	0.663
Highest single agent, ex23 singles	served	0.248	21	42	0	0	0.333	0.776
Bliss, ex21 fits	software	0.171	25	38	0	0	0.397	0.741
Bliss, ex23 singles	software	0.171	25	38	0	0	0.397	0.846
Loewe, ex21 fits	software	0.171	15	1	37	10	0.825	0.952
Highest single agent, ex21 fits	software	0.171	25	38	0	0	0.397	0.637
Highest single agent, ex23 singles	software	0.171	25	38	0	0	0.397	0.741

Table 7: Fitness of the combinations that grew, observed as the mean of the grown wells. The mean of observed minus predicted is signed: Bliss is high on these combinations and Loewe is low on them. Best Spearman and lowest RMSE within each call in bold. The software call changes which rule wins the fitness comparison, which is why claim 1 is reported under both.

Reference rule	Growth call	Grown combinations (n)	Spearman	RMSE (fitness)	Mean observed minus predicted (fitness)
Bliss, ex21 fits	served	21	0.717	0.259	-0.167
Bliss, ex23 singles	served	21	0.573	0.235	-0.142
Loewe, ex21 fits	served	21	0.713	0.208	+0.128
Highest single agent, ex21 fits	served	21	0.633	0.267	-0.183
Highest single agent, ex23 singles	served	21	0.476	0.243	-0.162
Bliss, ex21 fits	software	25	0.525	0.238	-0.118
Bliss, ex23 singles	software	25	0.865	0.137	-0.097
Loewe, ex21 fits	software	25	0.646	0.279	+0.185
Highest single agent, ex21 fits	software	25	0.431	0.252	-0.149
Highest single agent, ex23 singles	software	25	0.824	0.198	-0.158

The pairs. Twelve of the 15 pairs deviate below Bliss with a bootstrap interval excluding zero, two are additive and one is antagonistic (Table 8, Figure 3). The largest deviation is formic acid with lactic acid, which grew in no well under either call (−0.978 served). The two additive pairs are 5-HMF with acetic acid and 5-HMF with formic acid, under both calls. The one antagonistic pair, 5-HMF with furfural at +0.058 [+0.018, +0.103] served, flips to synergy under the software call at −0.040 [−0.052, −0.026], and the two 5-HMF pairs whose wells the served call reads as no growth

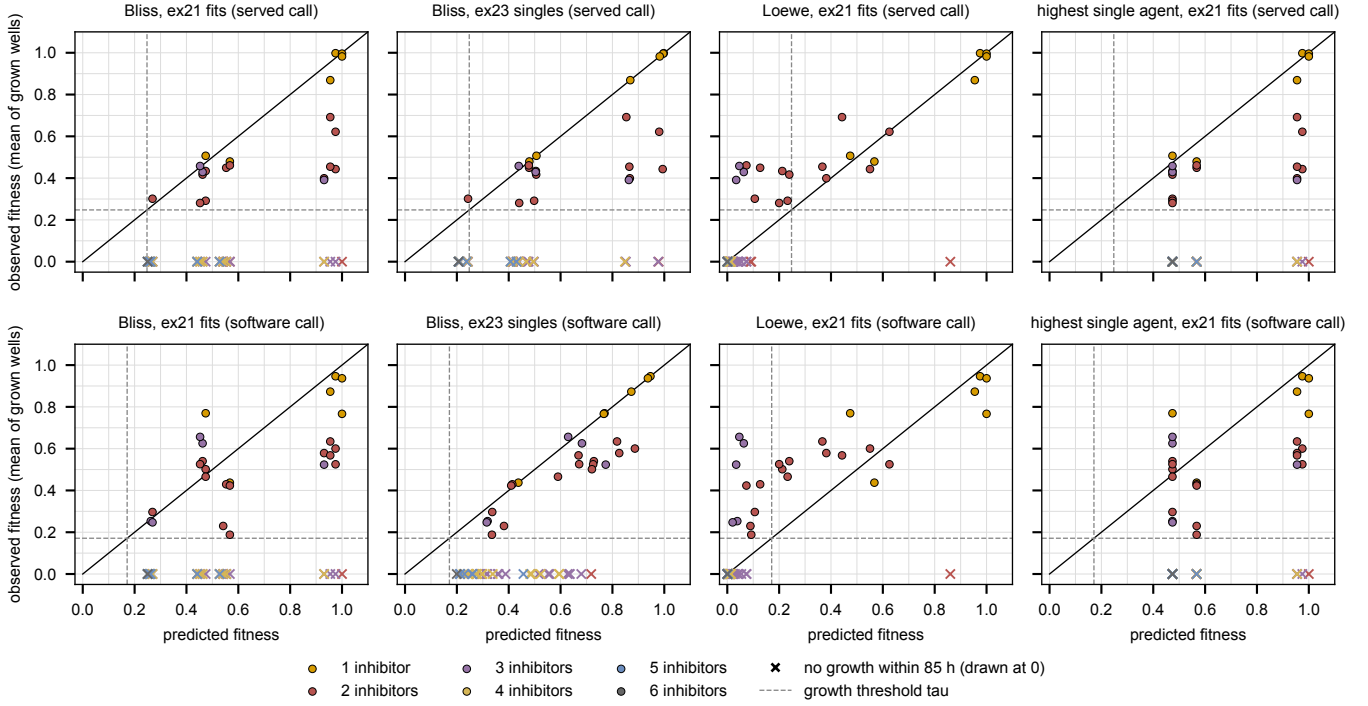


Figure 2: Observed against predicted fitness for the 63 combinations, by rule and growth call. Columns are the reference rules of Section 3.2 and rows are the two growth calls. Crosses are combinations with no growth within 85 h, drawn at 0, and the dashed lines mark that call's τ . The Bliss and highest-single-agent columns place nearly every combination above τ , which is the accuracy 0.333 of Table 6 seen directly; the Loewe column puts the no-growth set below τ and the grown set above its own prediction.

shrink by a factor of 3 under the software call. Every 5-HMF statement in this paragraph is therefore call dependent, and that is the same 12 wells of Section 2.

Table 8: The 15 pairs against Bliss independence from the observed ex23 singles, ordered by the served deviation. Observed is the mean of three wells with a well that did not grow counted as 0. The interval is 95 percent from 2000 resamples of the replicate wells of the pair and of both singles, and is shown for the served call; the call is synergy when the interval lies below zero, antagonism when above, additive otherwise. The software deviations carry their own intervals in `results/pair\_deviation.csv`.

Pair	Doses (g L ⁻¹)	Observed (fitness)	Bliss (fitness)	Deviation served	95% interval	Call, served	Deviation, software	Call, software
formic acid + lactic acid	1.00 / 20.00	0.000	0.978	-0.978	[-1.023, -0.934]	synergy	-0.718	synergy
acetic acid + formic acid	2.00 / 1.00	0.443	0.994	-0.551	[-0.596, -0.500]	synergy	-0.286	synergy
5-HMF + lactic acid	2.52 / 20.00	0.000	0.471	-0.471	[-0.497, -0.447]	synergy	-0.147	synergy
acetic acid + levulinic acid	2.00 / 6.00	0.399	0.867	-0.468	[-0.613, -0.358]	synergy	-0.247	synergy
5-HMF + levulinic acid	2.52 / 6.00	0.000	0.417	-0.417	[-0.434, -0.400]	synergy	-0.152	synergy
formic acid + levulinic acid	1.00 / 6.00	0.455	0.865	-0.411	[-0.555, -0.312]	synergy	-0.183	synergy
acetic acid + lactic acid	2.00 / 20.00	0.622	0.980	-0.359	[-0.475, -0.234]	synergy	-0.201	synergy
furfural + lactic acid	1.50 / 20.00	0.292	0.498	-0.206	[-0.254, -0.155]	synergy	-0.124	synergy
lactic acid + levulinic acid	20.00 / 6.00	0.692	0.854	-0.162	[-0.231, -0.097]	synergy	-0.101	synergy
furfural + levulinic acid	1.50 / 6.00	0.281	0.440	-0.160	[-0.204, -0.113]	synergy	-0.146	synergy
acetic acid + furfural	2.00 / 1.50	0.417	0.506	-0.089	[-0.139, -0.038]	synergy	-0.188	synergy
formic acid + furfural	1.00 / 1.50	0.434	0.505	-0.071	[-0.122, -0.017]	synergy	-0.219	synergy
5-HMF + acetic acid	2.52 / 2.00	0.449	0.479	-0.029	[-0.081, 0.015]	additive	+0.015	additive
5-HMF + formic acid	2.52 / 1.00	0.461	0.478	-0.017	[-0.057, 0.023]	additive	+0.014	additive
5-HMF + furfural	2.52 / 1.50	0.301	0.243	+0.058	[0.018, 0.103]	antagonism	-0.040	synergy

The isoboles. The three dose grids put the same question on a surface rather than at a point (Table 9, Figure 4). Furfural with acetic acid lies below Bliss by 0.082 [-0.142, -0.017] over its 81 interior cells, and formic acid with acetic acid by 0.446 [-0.536, -0.367], with 31 interior cells where Bliss predicts growth and nothing grew. The 5-HMF grid is additive against Bliss, -0.017 [-0.043, +0.007], and is flagged rather than trusted: one run, with growth islands at 5-HMF 2.0 g L⁻¹. No interior cell of any grid grew less than Loewe predicts, while 10, 3 and 12 cells grew where Loewe predicts none, so the observed front sits between the references and closest to Loewe, the same ordering the 63 combinations give.

Claim 1 is therefore supported in the direction registered, synergy relative to Bliss, with one correction to the registration: the number of grown combinations is 21 under the primary served call rather than the 25 the claim names, which is the software call's count. Hypothesis (untested): the steep single-agent curves, four of six at or near the slope bound,

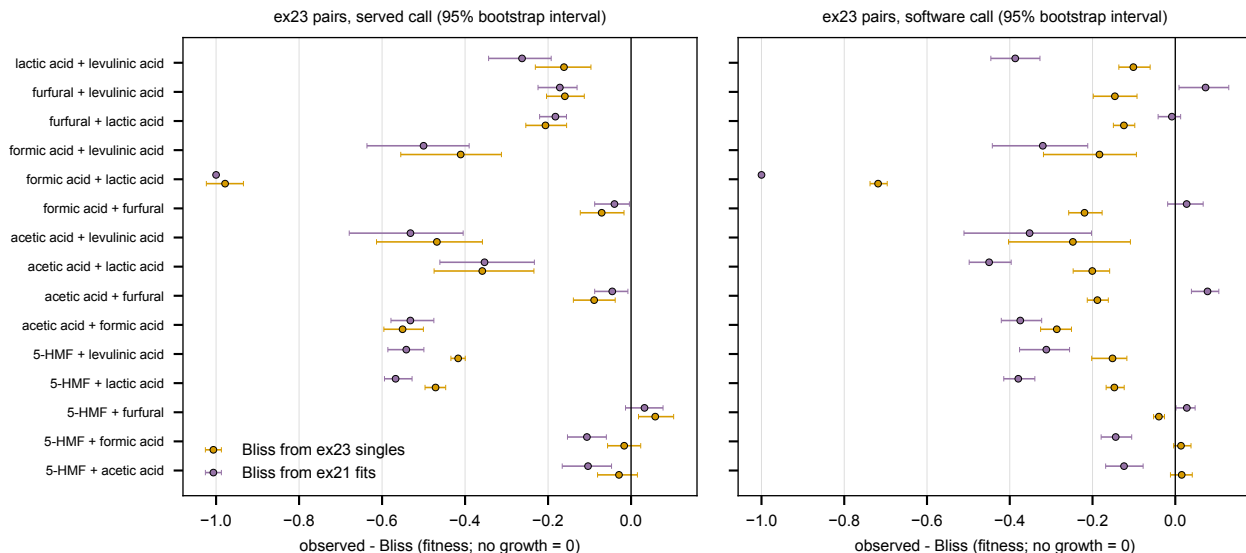


Figure 3: Observed minus Bliss for the 15 pairs, under both growth calls. Bars are 95 percent bootstrap intervals over resampled replicate wells. Orange takes Bliss from the observed ex23 singles, purple from the ex21 Hill fits with the fit-draw uncertainty included, which is why the purple intervals are wider. A bar whose interval clears zero is a synergy or antagonism call in Table 8; the 5-HMF pairs are the ones that move between the two calls.

Table 9: The three isobole grids, served call. Excess over Bliss is computed against the empirical Bliss surface built from each grid’s own axes, averaged over the 81 interior cells, with a 95 percent interval from 2000 resamples of the two plate wells per cell. The last three columns count interior cells that contradict a reference in one direction only. ex28 is analyzed and flagged, not trusted.

Run and pair	Interior cells (n)	Grew (n)	Mean excess over Bliss (fitness)	95% interval	Call against Bliss	Bliss grows, none grew (n)	Loewe grows, none grew (n)	Loewe none, growth seen (n)
ex26, furfural x acetic acid	81	14	-0.082	[-0.142, -0.017]	synergy	1	0	10
ex27, formic acid x acetic acid	81	24	-0.446	[-0.536, -0.367]	synergy	31	0	3
ex28, 5-HMF x acetic acid (flagged)	81	17	-0.017	[-0.043, 0.007]	additive	3	0	12

are what drive the Bliss and Loewe predictions this far apart. No calculation here separates that from the alternatives.

4.2 Claim 2 fails as registered: profile similarity does not anticipate which pairs deviate x

The registered expectation was that two inhibitors with similar deletion profiles, and so with shared targets, would be additive, and dissimilar ones would deviate. With the measured profiles in the matrix nothing in Table 10 supports it: the largest association is $+0.311$ at $p = 0.26$, every permutation p is above 0.26, and the sign flips between the centered Spearman and the top-100 Jaccard measure. Only 6 of the 15 pairs have two predicted profiles in that matrix, so the test is thin as well.

The only significant association is in the all-predicted matrix and has the opposite sign to the registration: the more similar two predicted profiles, the more synergistic the pair (raw Spearman -0.686 , $p = 0.006$ under the software call; -0.493 , $p = 0.066$ served). It is a chemistry-class regularity rather than a chemogenomic one. The three small acids have near-duplicate predicted profiles, centered Spearman 0.91 to 0.99, which is fingerprint similarity carried through a ridge fit whose 32 compounds include no C1 to C3 acid, and they are also the strongly synergistic pairs: formic with lactic acid grew in no well, acetic with formic acid deviates by -0.551 , acetic with levulinic acid by -0.468 . The association therefore says that weak-acid pairs synergize, which the fingerprint already encodes, and it depends on the growth call ($p = 0.07$ to $p = 0.43$ served). The three isoboles cannot separate the two readings: furfural with acetic acid has similarity 0.04 in the best-available matrix and 0.51 in the all-predicted one and is mildly synergistic, while formic with acetic acid has -0.08 and 0.99 and is strongly synergistic.

4.3 Claim 3 holds on public data: deletion profiles compose as the mean of the singles x

Over Hillenmeyer’s 26 two-compound environments the mean of the two matched single profiles has the highest R^2 of the three fixed rules in 24 of 26 pairs, and in 20 of the 22 that are not flagged (Table 11, Figure 7). The sum wins once and the maximum once, in a pair where every rule is below 0.14. Medians over the 26 are R^2 0.365 for the mean, 0.091 for the sum and 0.050 for the maximum, against 0.458 for the per-pair linear fit, whose median coefficients are $a = 0.529$

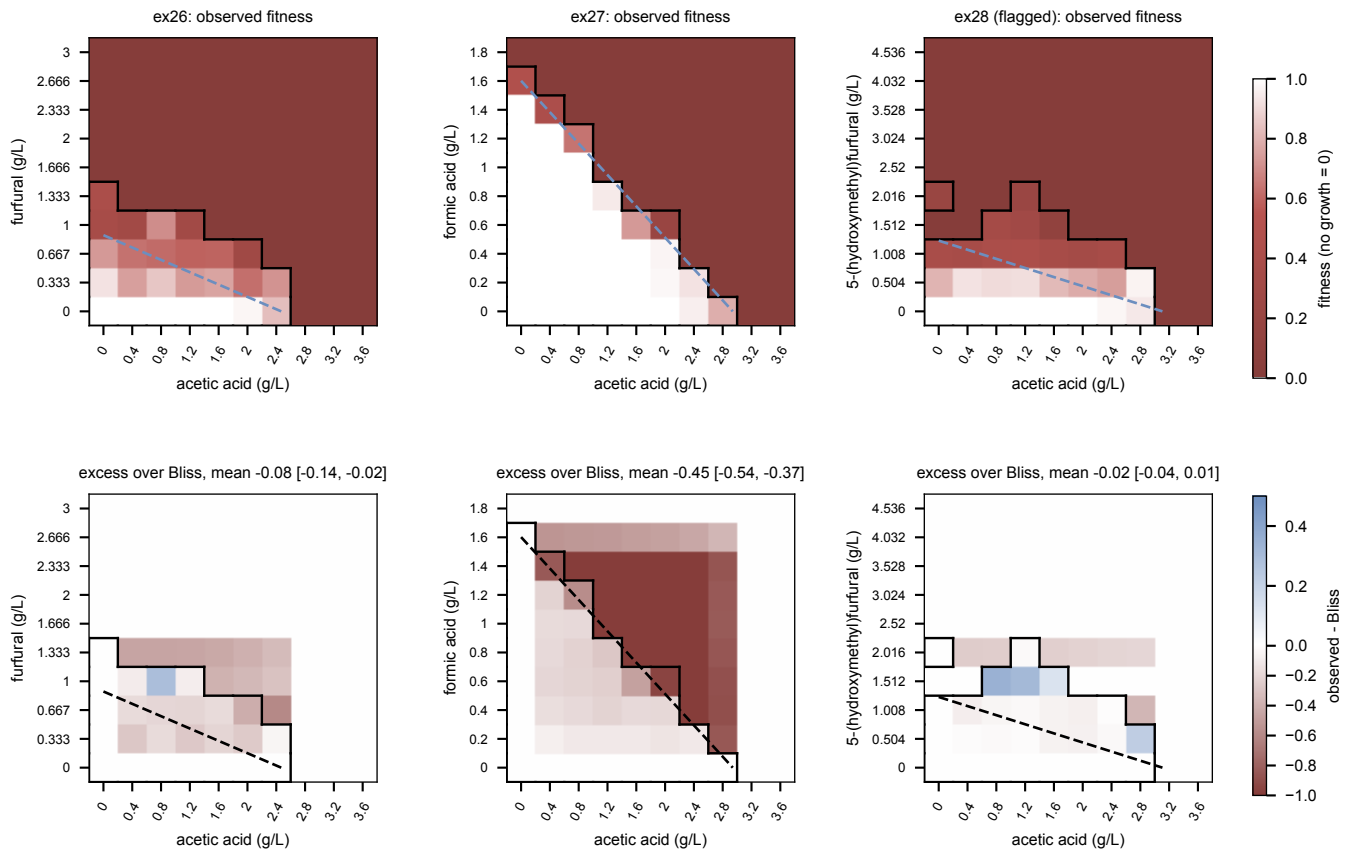


Figure 4: The three isobole grids, observed and against Bliss. Top row: observed fitness per dose cell, mean of two plates with a cell that did not grow at 0, the black contour around the cells that grew and the dashed blue line the Loewe additive front at that run's τ from the grid-axis Hill fits. Bottom row: observed minus the empirical Bliss surface, so blue is growth below independence. The formic acid grid is where the two references separate most: its observed front follows the Loewe line while 31 interior cells that Bliss expects to grow did not.

Table 10: Profile similarity against the pair deviation from Bliss, over the 15 pairs. A negative Spearman means similar profiles deviate further below Bliss, which is the opposite of the registered direction. Permutation p is over 10,000 draws and is set in bold below 0.05. The last column counts the pairs whose two profiles are both predictions, which is where the fingerprint regularity of the small acids enters.

Profile matrix	Similarity measure	Growth call	Pairs (n)	Spearman against deviation	Permutation p	Leave-one-pair-out range (Spearman)	Both profiles predicted (n)
best available	centered Spearman	served	15	+0.204	0.462	[+0.081, +0.481]	6
best available	centered Spearman	software	15	+0.311	0.263	[+0.226, +0.613]	6
best available	raw Spearman	served	15	-0.179	0.522	[-0.319, +0.011]	6
best available	raw Spearman	software	15	+0.086	0.768	[+0.007, +0.336]	6
best available	top-100 Jaccard, centered	served	15	-0.269	0.328	[-0.426, -0.099]	6
best available	top-100 Jaccard, centered	software	15	-0.022	0.940	[-0.113, +0.205]	6
all predicted	centered Spearman	served	15	-0.225	0.425	[-0.354, -0.051]	15
all predicted	centered Spearman	software	15	-0.525	0.050	[-0.662, -0.420]	15
all predicted	raw Spearman	served	15	-0.493	0.066	[-0.653, -0.380]	15
all predicted	raw Spearman	software	15	-0.686	0.006	[-0.771, -0.618]	15
all predicted	top-100 Jaccard, centered	served	15	-0.318	0.246	[-0.462, -0.166]	15
all predicted	top-100 Jaccard, centered	software	15	-0.709	0.005	[-0.780, -0.648]	15

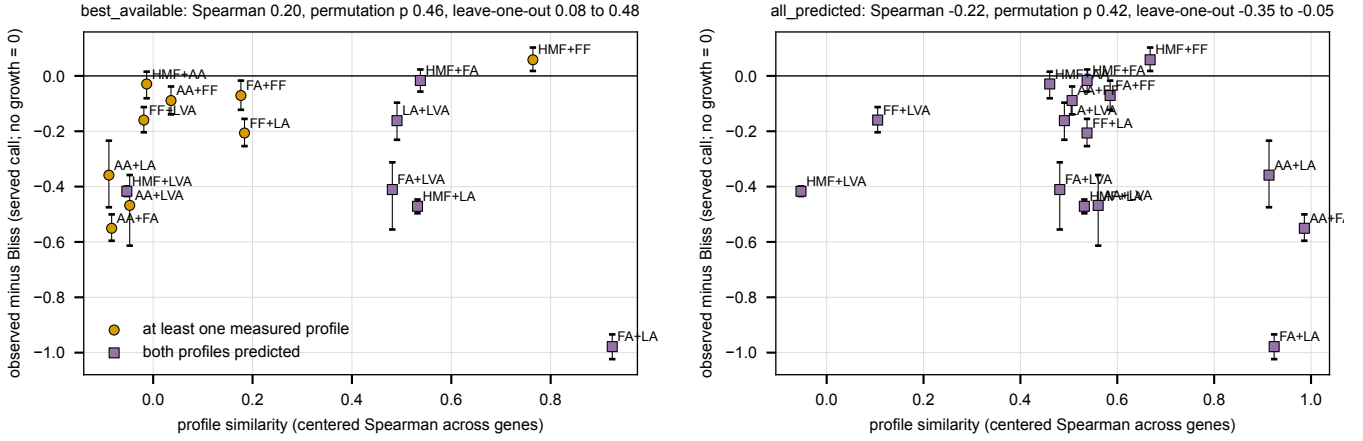


Figure 5: Pair deviation from Bliss against profile similarity, by matrix and growth call. One point per pair. The best-available panels carry the measured furfural and Hoepfner acetate columns and show no association; the all-predicted panels show the negative trend of Table 10, which the small-acid pairs at high similarity and large negative deviation produce.

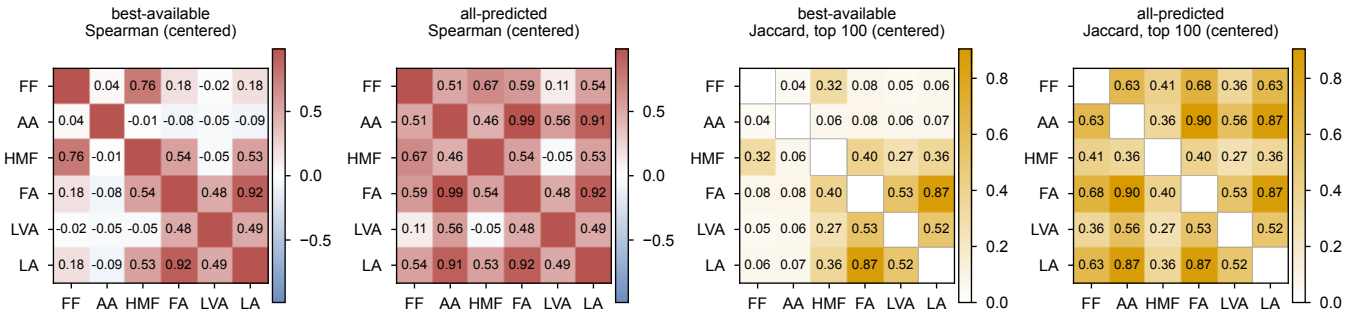


Figure 6: Centered profile similarity among the six inhibitors, in both matrices. Spearman correlation on a diverging scale and top-100 Jaccard overlap on a sequential scale, over the genes each pair of profiles shares. In the all-predicted matrix the three small acids form a near-duplicate block at 0.91 to 0.99. In the best-available matrix the Hoepfner acetate column is near zero against everything, at most 0.09 in absolute value, which is the measured cross-screen column failing to resemble any prediction.

and $b = 0.593$ with $a + b = 1.06$ and an intercept near zero. The finding is about magnitude and not about ordering: the sum and the mean order genes identically and share one Spearman of 0.579, so the sum's low R^2 is entirely its overshoot.

The matched singles do carry the pair. That Spearman of 0.579 compares with 0.066 for two random YPD singles and 0.114 for singles matched on the control set, and exceeds the matched null in 24 of 26 pairs at $p \leq 0.05$ over 200 draws (Table 12); both exceptions are pairs whose azole single sits a twofold step from the pair dose.

Table 11: How a two-compound profile relates to its two single profiles, over Hillenmeyer's 26 two-compound environments and the 22 that are not flagged. R^2 is of the rule as it stands, with no refit, so a rule of the wrong magnitude scores low or negative; Spearman is scale-free, which is why the sum and the mean share one value. The sum is the Bliss product on this scale (Equation 6). Best value per column in bold. The free linear fit is fitted per pair and so is an upper reference, not a rule that could be applied to an unmeasured pair.

Composition rule	Median R^2 26 pairs	Median R^2 22 unflagged	Median Spearman 26 pairs	Median Spearman 22 unflagged	Highest R^2 in
Sum (Bliss)	0.091	0.119	0.579	0.587	1 of 26
Mean	0.365	0.417	0.579	0.587	24 of 26
Max	0.050	0.106	0.411	0.482	1 of 26
Free linear	0.458	0.469	0.634	0.595	fitted per pair

Emergent and masked genes sit inside the noise. Pooled over the 26 pairs, 6,234 of 16,172 pair hits are emergent (0.385) and 18,341 of 28,279 single-union hits are masked (0.649), and the median emergent fraction of 0.355 is well below both nulls, 0.621 random and 0.613 control-set matched, in 24 of 26 pairs. The rate cannot be read as biology, because the only measurable noise floor is of the same size: the two near-replicate single environments in the release agree at Spearman 0.242 and 0.103, and 54 and 64 percent of one member's hits are not hits in the other (Table 12). With a $|z| > 2$ cut on single arrays, emergent and masked counts are dominated by irreproducibility, and these data

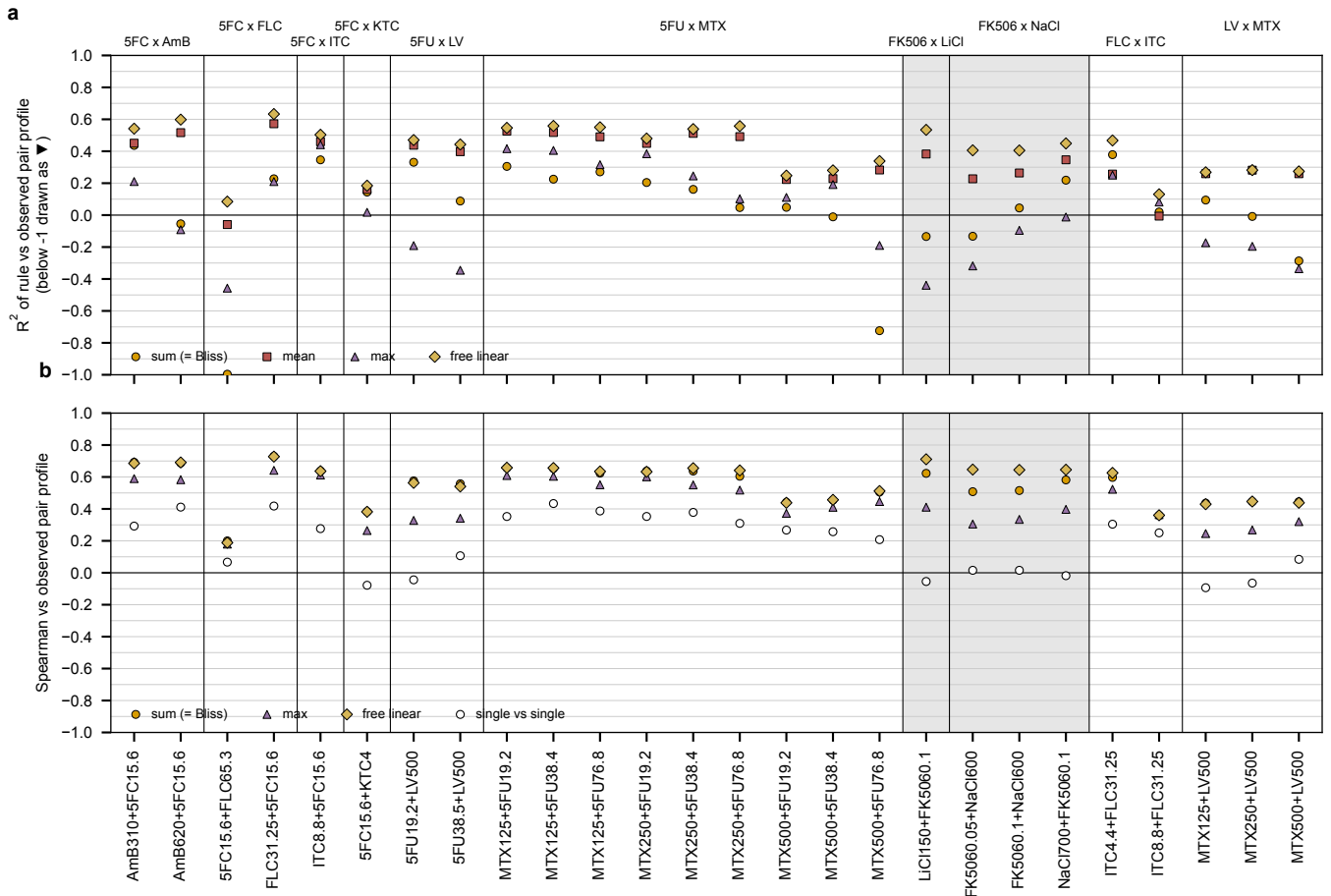


Figure 7: (a) R^2 of each composition rule against the observed pair profile, one group of bars per two-compound environment, grouped into pair families by the vertical rules; an R^2 below -1 is drawn as a down triangle at the floor. The mean is highest in all but two groups. (b) The same comparison by Spearman, where the sum and the mean coincide, with open circles giving the Spearman between the two matched single profiles, median 0.254. Gray bars are the four flagged tacrolimus pairs, whose nearest dosed single is 402 to 804-fold away. 5,677 to 5,704 genes per pair.

do not separate the two. The floor itself carries two caveats: $n = 2$, and the near-replicates span different control sets while every pair shares its control set with its singles, so the like-for-like floor is not measured.

Both fractions fall with the similarity of the two singles, Spearman -0.657 ($p = 2.6 \times 10^{-4}$) for emergent and -0.786 ($p = 2.0 \times 10^{-6}$) for masked over 26 pairs (Figure 8). Hypothesis (untested): two similar singles share hits, so their union covers more of the pair's hits and leaves fewer to be masked, which would produce this trend with no interaction at all.

Table 12: Hit-level composition against the two nulls, and the noise floor that bounds what it can mean. A hit is robust $|z| > 2$ within an environment. The nulls replace the matched singles by random YPD singles, 200 draws, the second restricted to singles sharing a control set with the pair. The last two rows are the only near-replicate single environments in the release, two 5-fluorouracil doses within 1 percent; their hit discordance exceeds the median emergent fraction above.

Quantity	Median observed	Median, random singles	Median, matched singles	Pairs beyond matched null ($p \leq 0.05$)	Pooled, 26 pairs
Emergent fraction (of pair hits)	0.355	0.621	0.613	24 of 26	0.385
Masked fraction (of single-union hits)	0.623				0.649
Sum-rule Spearman against the pair profile	0.579	0.066	0.114	24 of 26	
Near-replicate singles, 38.4 against 38.5 μM 5-fluorouracil	0.242			hits not shared	0.603 / 0.536
Near-replicate singles, 76.9 against 76.8 μM 5-fluorouracil	0.103			hits not shared	0.483 / 0.640

The weights move with dose. On the one dose grid HET provides, methotrexate against 5-fluorouracil at three doses each, the mean is the best fixed rule at all nine dose pairs, R^2 0.223 to 0.526, while the sum ranges from -0.724

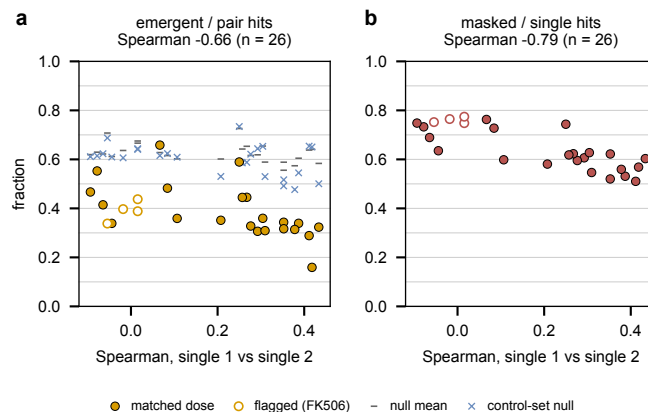


Figure 8: (a) Emergent fraction against the similarity of the two single profiles, one point per two-compound environment, with the two null means drawn as dashes and crosses at each pair’s own similarity. Every observed point lies below both nulls. (b) Masked fraction against the same similarity, falling more steeply, Spearman -0.786 . Open symbols are the four flagged tacrolimus pairs.

to 0.305 (Figure 9). The free-linear coefficients are not stable across the grid: a runs 0.128 to 0.757 and b 0.239 to 0.765, with the methotrexate weight high at methotrexate 500 μM and the 5-fluorouracil weight dominating at 125 μM . So the mean is a default rather than a constant. Hypothesis (untested): the weight follows whichever partner is relatively stronger at that dose ratio; no single-agent potency was computed on these environments to test it.

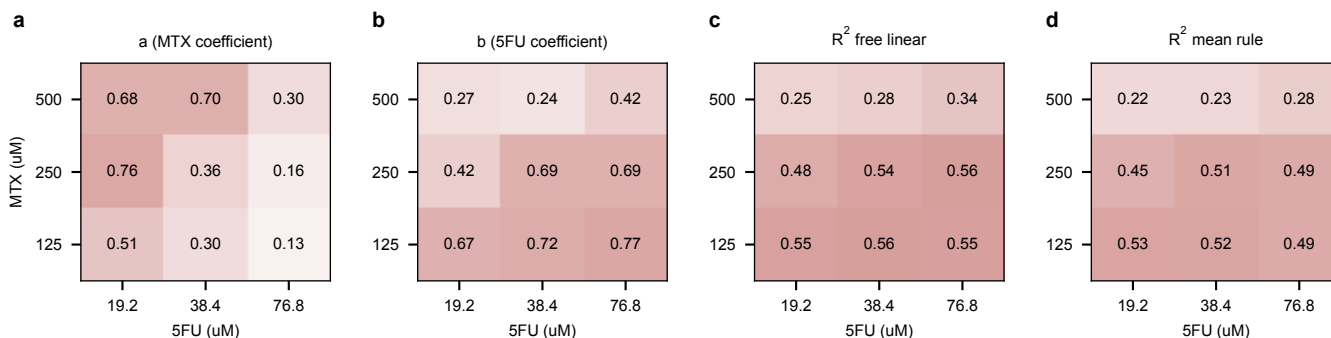


Figure 9: The methotrexate by 5-fluorouracil dose grid, nine dose pairs. (a) The free-linear weight a on the methotrexate single, which rises with the methotrexate dose. (b) The weight b on the 5-fluorouracil single, which dominates at the lower methotrexate dose. (c) R^2 of the free linear fit, 0.247 to 0.559, lowest at methotrexate 500 μM . (d) R^2 of the fixed mean rule, 0.223 to 0.526, which is below the free fit at every cell and above the sum at every cell.

Carrying the mean rule to the six inhibitors is a labeled hypothesis (untested). HET is a heterozygous diploid pool in aerobic YPD read as $\log_2$ depletion over 20 generations, the inhibitor profiles are haploid homozygous deletions in anaerobic SYN3 at IC30, and no two-compound inhibitor screen exists to check it. The rule is also a statement about the shape of a deletion profile, not about the host’s growth, which is what Section 4.1 scores.

4.4 Claim 4, post-analysis: the predicted profiles carry only the generic stress component $\times$

What the predictions are worth. Leave-one-compound-out over the 32 corrected compounds gives a median centered Spearman of 0.387, interquartile range 0.241 to 0.482, which is above experiment 038’s bar of 0.359 with 31 rather than 25 to 26 compounds fitted, and 038’s own fold means for these two compounds, 0.357 and 0.392, reproduce within that spread (Table 13, Figure 10). The two compound-level numbers are not equally meaningful. Furfural reaches 0.341 against a profile whose replicate reliability is 0.769. 5-HMF reaches 0.425 against a profile whose replicate reliability is -0.048 , so that score is agreement with noise plus the shared stress component and is not evidence that 5-HMF biology is predicted.

Acetate shows the limit from the other side. The predicted acetic acid profile correlates with Hoepfner’s measured

sodium acetate profile at -0.084 over 3,453 genes, and the two measured acetate profiles, Vanacloig build 001 and Hoepfner, do not agree with each other either, -0.011 centered. The acetic acid and sodium acetate predictions are nearly the same profile, 0.932 centered, so the salt-for-acid substitution of Section 2 changes little on the prediction side and the disagreement is between screens; which screen bAID in YPD resembles is untested. Build 001 and build 002 give the same furfural profile, raw Spearman 0.99998 , so the normalization change between builds separates nothing here.

Table 13: Compound-cold profile checks. Centering subtracts each gene’s mean over the 32 build-002 compounds, which removes the component every compound shares, so the centered column is the compound-specific agreement and the raw column is inflated by that shared part. A score against a profile whose replicate reliability is near or below zero is a score against noise.

Comparison	Genes (n)	Centered Spearman	IQR (Spearman)	Raw Spearman	Measured replicate reliability (Spearman)
Leave-one-compound-out over 32 compounds, median	32 compounds	0.387	[0.241, 0.482]		
furfural, leave-one-out against build 002	3492	0.341		0.258	0.769
5-HMF, leave-one-out against build 002	3503	0.425		0.260	-0.048
acetic acid predicted against Hoepfner sodium acetate	3453	-0.084		0.043	
acetic acid prediction against sodium acetate prediction	3587	0.932		0.990	
levulinic acid predicted against build 001	3485	0.190		0.255	0.147
Vanacloig build 001 sodium acetate against Hoepfner sodium acetate	3376	-0.011		0.034	
furfural, build 001 against build 002	3492	0.998		1.000	0.769

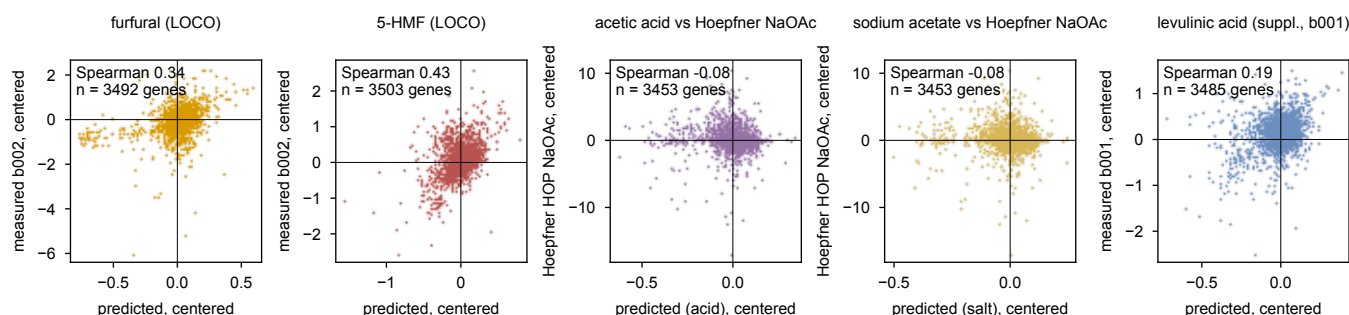


Figure 10: Predicted against measured centered profiles, one point per deletion strain. Panels are furfural and 5-HMF with themselves held out of the fit, acetic acid against the Hoepfner sodium acetate profile, and the supplementary build-001 levulinic acid row. The furfural panel is the one comparison whose measured axis has a replicate reliability above 0.5; the acetate panel is the cross-screen comparison that fails, and its scatter is round rather than elongated.

The enrichment test, against the control that matters. A nominated set is enriched for something in every case, so the test is whether it is enriched for anything the 32-compound gene mean does not already give (Table 14, Figure 11). It is not. The measured furfural profile’s top 100 genes carry no term at FDR 0.05, the best being ribosome at 0.057. The predicted profiles carry terms, but control terms: 19 of 20 for furfural, 40 of 43 for levulinic acid, 48 of 49 for acetic acid. The 11 terms the acetic acid prediction shares with the Hoepfner profile are all in the control too, vesicle transport and vesicle tethering complex among them. On centered profiles no compound shares a single significant term between measured and predicted.

Table 14: GO terms at FDR 0.05 over the top 100 most sensitive genes, for each compound with a measured profile and for its compound-cold prediction. The control is the build-002 gene mean over the 32 compounds, which is what a model with no compound-specific signal outputs; the control columns are computed on raw profiles only. A term in the predicted-and-control column is not evidence about the compound.

Measured profile	Target	Shared top genes (n)	Terms, measured (n)	Terms, predicted (n)	Shared terms (n)	Terms, control (n)	Measured and control (n)	Predicted and control (n)
furfural, build 002	raw	19	0	20	0	58	0	19
5-HMF, build 002 (noise)	raw	22	0	4	0	58	0	4
acetic acid, Hoepfner	raw	10	20	49	11	58	11	48
levulinic acid, build 001	raw	24	0	43	0	58	0	40
furfural, build 002	centered	15	0	13	0			
5-HMF, build 002 (noise)	centered	43	15	6	0			
acetic acid, Hoepfner	centered	15	5	5	0			
levulinic acid, build 001	centered	38	11	0	0			

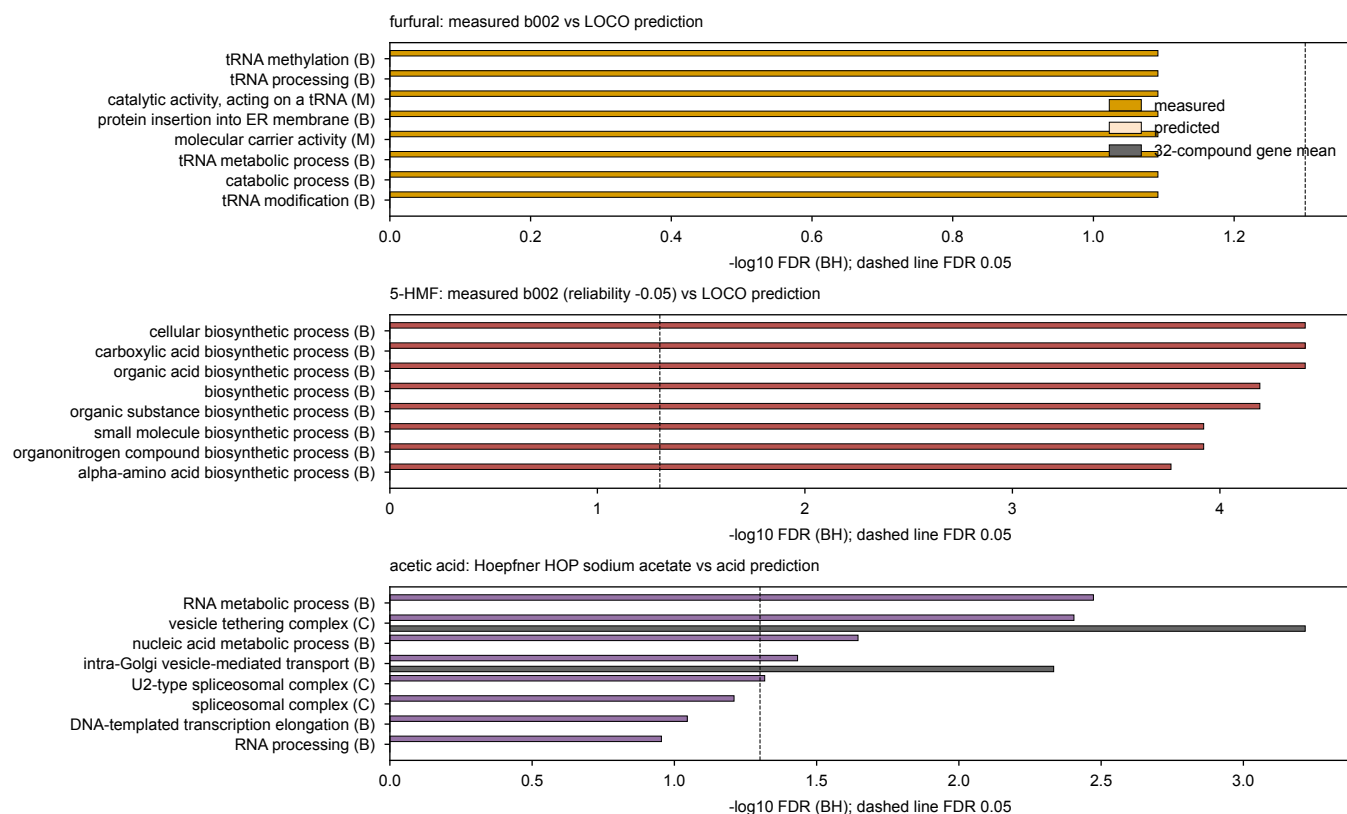


Figure 11: The eight strongest terms of each measured centered profile, beside the prediction and the control. Bars are $-\log_{10}$ of the Benjamini-Hochberg FDR for the measured profile, for its compound-cold prediction, and for the 32-compound gene mean; a zero bar means the term was neither significant nor among that set's ten best. No compound has a term where the measured and predicted bars are both significant.

The nominations, which are hypotheses. Composing the six single columns by the sum rule and categorizing gives 60 to 68 core genes per combination in the best-available matrix and 105 in the all-predicted one (Table 15). The core set is led by Golgi-transport and chromatin genes, *YPT6*, *RIC1*, *VPS63*, *COG5*, *GOS1*, *SNF6* and *DEG1*, which are the genes sick under most Vanacloig compounds. Conflict genes, sensitive in one member and resistant in another, number 5 to 25 per combination in the best-available matrix and exactly 1 in the all-predicted one, because predicted profiles are too alike to disagree. No screen of the bAID pool under any combination exists, so every one of these is a hypothesis for an experiment that has not been run, and a nominated set's enrichment is not evidence of compound biology unless it exceeds the control above.

Table 15: Nominated gene counts over the 35 combinations of two or three inhibitors, rule sum. The core set is defined on the six single columns and so varies only with which genes every member carries; Hoepfner covers 3,453 of the 3,587 background genes, which is why the best-available core set is smaller. The composed top 100 is 100 by construction.

Category	Median genes, best available (n)	Range, best available (n)	Median genes, all predicted (n)	Range, all predicted (n)
Core, sensitive in at least four of six singles	67	60 to 68	105	105 to 105
Compound specific, sensitive in exactly one member	65	7 to 125	22	2 to 64
Conflict, sensitive in one member and resistant in another	11	5 to 25	1	1 to 1
Composed top 100	100	100 to 100	100	100 to 100

5 What this means for the model, and for the figure x

5.1 The bars are model-free and they are already on file x

The wet-lab data are a clean validation target, and the reference rules set the level a trained model has to reach before it has said anything (Table 16). Growth over the 63 combinations is separated by Loewe additivity at AUROC 0.958, with one false positive, and the ordering of the 21 grown combinations is recovered by Bliss from the observed singles at

Spearman 0.573. Both come from the single-agent curves alone, so a model that reaches them has matched something that needs no training data beyond the six titrations in the same run.

Table 16: The bars a dose- and mixture-aware model has to clear on these data, served call, taken from the best reference rule for each test. The two growth rows and the two fitness rows differ only in whether the six single-agent combinations are included, and the fitness rows are the harder test: 15 grown combinations of two or more inhibitors is a small sample and Bliss already orders them at 0.261.

Test	Metric	Best model-free rule	n	Value to beat
Growth or no growth, 63 combinations	AUROC	Loewe, ex21 fits	63	0.958
Growth or no growth, 57 combinations of two or more	AUROC	Loewe, ex21 fits	57	0.944
Fitness of the combinations that grew	Spearman	Bliss, ex23 singles	21	0.573
Fitness of the grown combinations of two or more	Spearman	Bliss, ex23 singles	15	0.261
Fitness of the combinations that grew	RMSE (fitness)	Loewe, ex21 fits	21	0.208

What the chemogenomic representation adds to that is, so far, nothing measurable. Compound-cold ridge profiles reach the published bar on the gene-level task (Section 4.4) and still carry only the generic stress component; profile similarity does not anticipate which pairs deviate once a measured profile is in the matrix (Section 4.2); and the one composition rule that is measurable is measurable on public data for compounds that are not these (Section 4.3). The honest preliminary claim is the model-free one, which is also why the run order put these four claims before any training.

5.2 The panel this fills, and what each cell needs $\times$

The case study occupies one figure of the manuscript, planned as four rows of three panels and expected to compress to three rows by merging the two middle ones (Table 17). Seven of the twelve cells are produced by a committed script today; two depend on a running job, and three need code that is not written.

Table 17: The planned panel, one row per cell. Status carries the figure-gate glyphs of the status key: on file (●) when a committed script already produces the panel, partly or running (◐) when part of it is on file or a submitted job will produce it, and to launch or to build (○) when the round is not submitted or the code does not exist. None of the three means approved. The bottom row is the claim the figure exists to make: whether a model trained on public screens can predict this host under these environment perturbations.

Row	Panel	Content	Source	Status
1 concept	a	The task: a compound enters the cell graph transformer as a token the genes attend to; the strain is read at its deleted genes, the host at the mean over an empty genotype	draw.io schematic of the 038 environment encoder and the 040 host head	○ to build
1	b	What is trained on: the pooled chemogenomic table beside the private Bioscreen data, 977 wells, six inhibitors, 63 combinations, three isoboles, one strain	033 store_against_plan.py; 040 wetlab_table.py	● on file
1	c	Evaluation: compound-cold folds and the per-compound centered Spearman against each compound’s reliability ceiling; the wet-lab test, growth over 63 and fitness over the grown, against Bliss and Loewe	038 protocol; 040 mixture_rules.py	● on file
2 benchmarks	d	The ladder on the corrected store: nested ridge on FCFP4 counts against kernels, nearest neighbors and twelve embeddings	038 round 1	● on file
2	e	The transformer against ridge: bilinear control, environment-encoder head, nine-seed ensembles, the stack, per compound	038 round 2	◐ running
2	f	Learning curve in fitted compounds, and the effect of adding further screens to the gene-level task	038 round 17; 040 round 1 source arms	○ to launch
3 environment	g	Dose: public IC30 against ex21 IC30 per shared compound, and the dose-aware encoder’s predicted single-agent curves against ex21	040 single_agent_curves.py; 040 round 1 dose arms	◐ partly
3	h	Mixture rule on public data: a pair profile is the mean of its singles, and the emergent gene rate sits inside the noise	040 het_mixture_composition.py	● on file
3	i	Predicted against measured profile for furfural and 5-HMF, for ridge and for the transformer encoder, with GO overlap against the gene-mean control	040 inhibitor_profiles.py	◐ partly
4 wet lab	j	The ex21 single-agent curves with the model’s predicted curves, trained on the public anchors only and calibrated on ex21, both shown	040 round 1	○ to launch
4	k	The 63 combinations: observed against Bliss, Loewe and the model, by number of inhibitors, with growth AUROC and fitness Spearman in the panel	040 mixture_rules.py for the rules; the model to launch	◐ partly
4	l	The isoboles: observed grid, Loewe front, the model’s predicted front, and the excess over Bliss	040 mixture_rules.py for the rules; the model to launch	◐ partly

If the model does not clear the bars of Table 16, panels j to l show the rules and the model side by side and the text says so, which is why the bottom row is drawn from the reference rules first.

5.3 Round 1, which has not run x

The adaptation keeps experiment 035 round 11’s recipe, the best on record: a compound token in front of the 6,607 gene tokens, one further layer, a one-layer encoder, prior weight zero, 50 epochs keeping the last, fit on the non-test pool and stacked with nested ridge. Three things change, each an ablation: a set of compound tokens, one per inhibitor in the medium, in place of the single token; a log-molar dose scalar modulating each token, with the public IC30 doses and dose zero as the anchors; and a host-growth head read from the mean of the cell-in-medium state and the compound tokens after they read the genome, the empty-genotype path, fit on the ex21 curves and scored on the 63 combinations and the isoboles against the references of Section 4.1. The gene-level head stays on the public task so the compound representation is shared.

Hypothesis (untested), and it is the only statement this section can make: the trainer is being written, no run exists, and nothing above comes from a model. On the record from experiment 035 the transformer encoder ties nested ridge on the gene-level task, and the compound-cold profiles of Section 4.4 are generic, so the expectation is that the encoder’s profiles behave the same way on claim 2 and that the host head has to earn its place from the dose anchors and the ex21 curves rather than from the profile representation. Neither half of that has been measured. The next number this document can carry is a growth AUROC and a fitness Spearman from a finished run, set beside the 0.958 and the 0.573 of Table 16.